The essential lipoprotein maturation enzyme Lgt is required for flagellar and membrane integrity in *Helicobacter pylori*

Tristan RUFFIOT, Bertrand LEGERET, Yonghua LI-BEISSON, Ivo G. BONECA and Nienke BUDDELMEIJER

Lipoproteins play an important role in physiology and virulence of bacteria. They are characterized by their lipid anchor allowing them to attach to the phospholipid membranes. The first step in lipoprotein maturation is catalyzed by the essential inner-membrane integral protein Lgt that transfers a diacylglyceryl group from phosphatidyglycerol to pre-pro-lipoproteins. Lgt is conserved in bacteria, absent in eukaryotes and its membrane orientation provides access to small molecule inhibitors. We hypothesize that Lgt is an interesting target for the development of antibiotics.

We confirmed the essentiality of Lgt on growth, viability and morphology of *H. pylori* using a Lgt depletion strain, where *lgt* is expressed from a regulated *cag*UT promoter from a neutral locus on the chromosome. Depletion of Lgt leads to an aberrant cell morphology; round, large cells with bulges and mislocalized flagella are observed by scanning electron microscopy. Interestingly, under prolonged depletion conditions cells compensate for the lack of Lgt and cells are able to regrow. These so-called revertants adapt a rod cell shape but are shorter than wild type cells. Transcriptome analyses showed that expression of two phospholipid biosynthesis genes and several flagellar genes are affected in the revertants, while *lgt* expression levels are slightly lower than wild type. Transmission electron microscopy revealed a reduced number of polar flagella with a disorganized flagellar sheath, however, cells are still motile as demonstrated by video microscopy and single cell tracking. Phospholipid analyses showed an increase in cardiolipin and decrease in phosphatidylglycerol during Lgt depletion, which was partially restored in the revertants. Together, our data reveal that absence of lipoprotein maturation leads to an envelope stress response that translates into adaptation of membrane and flagellar integrity and motility.